

COLLEGE PARK, MD, USA

WMO: 722244

Lat: 38.981N

Lon: 76.922W

Elev: 15

StdP: 101.14

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 00358

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.5	-7.1	-19.8	0.6	-6.7	-17.2	0.8	-3.9	8.9	1.9	8.1	1.5	2.4	240	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	35.1	23.5	33.3	23.1	32.3	22.8	25.5	30.8	24.9	30.1	24.3	29.4	2.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.1	27.1	23.4	18.2	26.6	22.8	17.6	26.2	78.2	31.4	75.5	30.2	73.1	29.3	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 7.5	6.1	5.1	DB	-12.5	36.9	1.8	2.0	-13.9	38.4	-14.9	39.5	-16.0	40.7	-17.3	42.1	(4)
(5)			WB	-13.2	26.4	2.0	0.8	-14.6	27.0	-15.8	27.4	-16.9	27.8	-18.3	28.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.2	1.9	3.3	7.4	13.8	18.8	23.6	26.0	24.9	21.2	15.0	8.4	4.8	(6)
(7)		DBStd	9.45	5.08	5.27	5.11	4.84	4.34	3.24	2.89	2.69	3.60	4.57	4.32	4.58	(7)
(8)		HDD10.0	848	257	196	114	17	0	0	0	0	0	8	80	175	(8)
(9)		HDD18.3	2332	510	420	341	153	51	2	0	0	12	126	297	421	(9)
(10)		CDD10.0	2363	6	9	33	129	272	408	497	463	337	162	33	13	(10)
(11)		CDD18.3	806	0	0	1	16	64	161	238	204	99	21	0	0	(11)
(12)		CDH23.3	8141	0	2	18	232	631	1640	2625	1973	851	166	3	0	(12)
(13)	CDH26.7	3289	0	0	1	76	203	653	1212	808	297	38	0	0	(13)	
(14)	Wind	WSAvg	1.6	2.0	2.1	2.2	2.1	1.5	1.4	1.3	1.1	1.2	1.4	1.6	1.7	(14)
(15)	Precipitation	PrecAvg	1085	78	71	91	80	109	89	105	112	97	82	87	86	(15)
(16)		PrecMax	1504	212	174	202	190	231	299	275	279	351	202	182	180	(16)
(17)		PrecMin	747	16	14	19	23	27	15	19	6	10	0	13	16	(17)
(18)		PrecStd	181	41	40	44	41	48	57	55	68	75	49	47	47	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.8	22.4	26.1	32.3	32.7	36.3	37.7	36.0	34.0	30.9	23.0	21.2	(19)
(20)			MCWB	15.9	15.4	15.1	18.2	22.2	23.3	23.5	23.7	22.7	21.8	15.8	17.6	(20)
(21)		2%	DB	16.3	17.9	22.5	28.8	31.1	33.7	35.9	33.9	32.4	27.3	20.6	17.8	(21)
(22)			MCWB	12.8	12.7	14.9	18.3	21.6	22.7	23.3	23.3	22.6	20.2	15.1	15.9	(22)
(23)		5%	DB	13.0	15.2	19.8	26.1	29.1	32.3	33.9	32.4	30.2	25.1	17.9	15.1	(23)
(24)			MCWB	9.0	9.6	13.7	17.0	20.7	22.5	23.7	23.1	21.8	19.0	13.4	12.1	(24)
(25)	10%	DB	10.1	11.7	17.0	22.8	27.3	30.8	32.4	31.1	27.9	22.6	16.1	12.2	(25)	
(26)		MCWB	6.9	7.0	11.0	15.4	19.5	21.9	23.2	22.6	20.8	17.9	12.0	9.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.1	16.8	17.3	21.2	23.8	25.5	26.6	26.5	25.1	23.4	18.2	18.6	(27)
(28)			MCD B	18.0	19.9	21.1	26.5	29.5	30.7	32.5	32.3	29.5	26.6	20.9	19.9	(28)
(29)		2%	WB	14.0	14.4	15.9	19.5	22.7	24.6	25.6	25.4	24.1	22.2	16.6	16.4	(29)
(30)			MCD B	16.3	18.3	20.8	24.9	28.3	30.5	31.3	30.2	29.0	25.5	19.1	18.4	(30)
(31)		5%	WB	9.9	10.1	14.1	18.1	21.9	23.9	25.1	24.7	23.5	21.0	14.8	13.0	(31)
(32)			MCD B	12.4	13.6	19.3	24.5	27.2	29.7	30.6	29.2	27.7	23.9	17.5	14.8	(32)
(33)	10%	WB	7.1	7.8	11.8	16.6	21.0	23.2	24.5	24.0	22.6	19.2	12.9	10.1	(33)	
(34)		MCD B	9.2	11.1	15.5	22.4	25.4	28.3	29.8	28.5	26.5	22.1	15.5	11.6	(34)	
(35)	Mean Daily Temperature Range	MDBR	8.8	9.5	10.6	12.1	11.2	11.1	10.7	10.5	10.7	11.0	10.4	8.5	(35)	
(36)		5% DB	MCD BR	13.6	14.7	15.3	16.4	14.1	12.9	12.4	12.1	13.1	13.0	13.7	11.7	(36)
(37)			MCWBR	10.5	9.6	9.2	7.1	6.0	4.2	3.4	3.9	4.4	6.5	8.4	9.8	(37)
(38)		5% WB	MCD BR	12.1	12.8	14.0	14.2	12.2	11.0	10.4	10.2	10.0	10.5	10.5	10.5	(38)
(39)			MCWBR	10.6	10.0	9.7	7.3	5.6	4.3	3.5	3.8	4.0	6.4	8.2	9.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.310	0.323	0.358	0.403	0.444	0.501	0.521	0.516	0.430	0.363	0.325	0.311	(40)	
(41)		taud	2.484	2.429	2.396	2.290	2.205	2.088	2.047	2.046	2.262	2.452	2.517	2.518	(41)	
(42)		Ebn at Noon	867	901	899	875	841	791	770	762	813	843	842	839	(42)	
(43)		Edh at Noon	79	95	109	129	143	161	166	163	123	92	76	72	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.07	2.92	3.87	4.92	5.54	6.12	6.02	5.36	4.36	3.26	2.36	1.81	(44)	
(45)		RadStd	0.17	0.33	0.28	0.38	0.63	0.47	0.31	0.29	0.51	0.36	0.20	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (73 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+124	+71

Nomenclature: See separate page